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Batch: 26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
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Mr. Bingi Raju			
Course Code	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week6 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number: 11.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question		Expected Time to complete
1	Lab 11: Data Structures with AI Implementing Fundamental Data Structures using AI Assistance Lab Objectives:		Week6 - Wednesday

	By the end of this lab, students will be able to: • Design and implement fundamental data structures in Python using AI assistance. • Effectively prompt AI tools (e.g., GitHub Copilot) for code generation, optimization, and documentation. • Understand and compare core data structures: Arrays, Linked Lists, Stacks, Queues, Priority Queues, Trees, and Graphs. • Improve code readability, efficiency, and maintainability using AI-generated suggestions. Learning Outcomes After completing this lab, students will be able to: • Apply appropriate data structures to solve real-world problems. • Analyze time and space complexity of different data structure operations. • Use AI tools responsibly to assist (not replace) logical thinking and problem-solving. • Validate, test, and refine AI-generated code.	
	Task 1: Smart Contact Manager (Arrays & Linked Lists) Scenario SR University's student club requires a simple Contact Manager Application to store members' names and phone numbers. The system should support efficient addition, searching, and deletion of contacts. Tasks 1. Implement the contact manager using arrays (lists). 2. Implement the same functionality using a linked list for dynamic memory allocation. 3. Implement the following operations in both approaches: ○ Add a contact ○ Search for a contact ○ Delete a contact 4. Use GitHub Copilot to assist in generating search and delete methods. 5. Compare array vs. linked list approaches with respect to: ○ Insertion efficiency ○ Deletion efficiency Expected Outcome • Two working implementations (array-based and linked-list-based). • A brief comparison explaining performance differences.	

WelcomeAAC 11.4.py X

C:\Users\shash> shash > Downloads > AAC 11.4.py > ArrayContactManager

```
1 class ArrayContactManager:
2     def __init__(self):
3         self.contacts = []
4     def add_contact(self, name, phone):
5         self.contacts.append({'name': name, 'phone': phone})
6     def search_contact(self, name):
7         for contact in self.contacts:
8             if contact['name'] == name:
9                 return contact['phone']
10        return None
11    def delete_contact(self, name):
12        for i, contact in enumerate(self.contacts):
13            if contact['name'] == name:
14                del self.contacts[i]
15            return True
16        return False
17
18 class Node:
19     def __init__(self, name, phone):
20         self.name = name
21         self.phone = phone
22         self.next = None
23
24 class LinkedListContactManager:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\Downloads> conda activate Shashidhar
PS C:\Users\shash\Downloads> & 'c:\Users\shash\anaconda3\envs\shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63770' '-' 'c:\Users\shash\Downloads\AAC 11.4.py'

Array search Alice: 123
Array delete Bob: True
Array search Bob after delete: None
Linked search Charlie: 789
Linked delete Dave: True
Linked search Dave after delete: None
PS C:\Users\shash\Downloads>

CHAT

CONTACT MANAGER IMPLEMENTATION WITH ARRAYS AND LINKED LI...

o Add a contact
o Search for a contact
o Delete a contact
4. Use GitHub Copilot to assist in generating search and delete methods.
5. Compare array vs. linked list approaches with respect to:
o Insertion efficiency
o Deletion efficiency
Expected Outcome
• Two working implementations (array-based and linked-list-based).
• A brief comparison explaining performance differences.
give code in python without comments

✓ pwd
✓ Created array_contact_manager.py
✓ Created linked_list_contact_manager.py
✓ Created comparison.txt

I've created three files in your current directory (C:\shash) with the implementations:
• array_contact_manager.py: Array-based contact manager
• linked_list_contact_manager.py: Linked list-based contact manager
• comparison.txt: Brief performance comparison

The code is provided without comments as requested. You can run these files directly in Python to test the functionality. For example, you could create a simple test script to instantiate the classes and call the methods.

+ AAC 11.4.py
Describe what to build next
Grok Code Fast

WelcomeAAC 11.4.py X

C:\Users\shash> shash > Downloads > AAC 11.4.py > ArrayContactManager

```
18 class Node:
19     def __init__(self, name, phone):
20
21         self.phone = phone
22         self.next = None
23
24     class LinkedListContactManager:
25         def __init__(self):
26             self.head = None
27         def add_contact(self, name, phone):
28             new_node = Node(name, phone)
29             if not self.head:
30                 self.head = new_node
31             else:
32                 current = self.head
33                 while current.next:
34                     current = current.next
35                 current.next = new_node
36         def search_contact(self, name):
37             current = self.head
38             while current:
39                 if current.name == name:
40                     return current.phone
41             current = current.next
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\Downloads> conda activate Shashidhar
PS C:\Users\shash\Downloads> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63770' '-' 'c:\Users\shash\Downloads\AAC 11.4.py'

Array search Alice: 123
Array delete Bob: True
Array search Bob after delete: None
Linked search Charlie: 789
Linked delete Dave: True
Linked search Dave after delete: None
PS C:\Users\shash\Downloads>

```

Welcome AAC 11.4.py X
C:\Users\shash\Downloads> AAC 11.4.py > ArrayContactManager
24 class LinkedListContactManager:
36     def search_contact(self, name):
40         return current.phone
41         current = current.next
42     return None
43     def delete_contact(self, name):
44         if not self.head:
45             return False
46         if self.head.name == name:
47             self.head = self.head.next
48             return True
49         current = self.head
50         while current.next:
51             if current.next.name == name:
52                 current.next = current.next.next
53                 return True
54             current = current.next
55         return False
56
57 array_cm = ArrayContactManager()
58 array_cm.add_contact("Alice", "123")
59 array_cm.add_contact("Bob", "456")
60 print("Array search Alice:", array_cm.search_contact("Alice"))

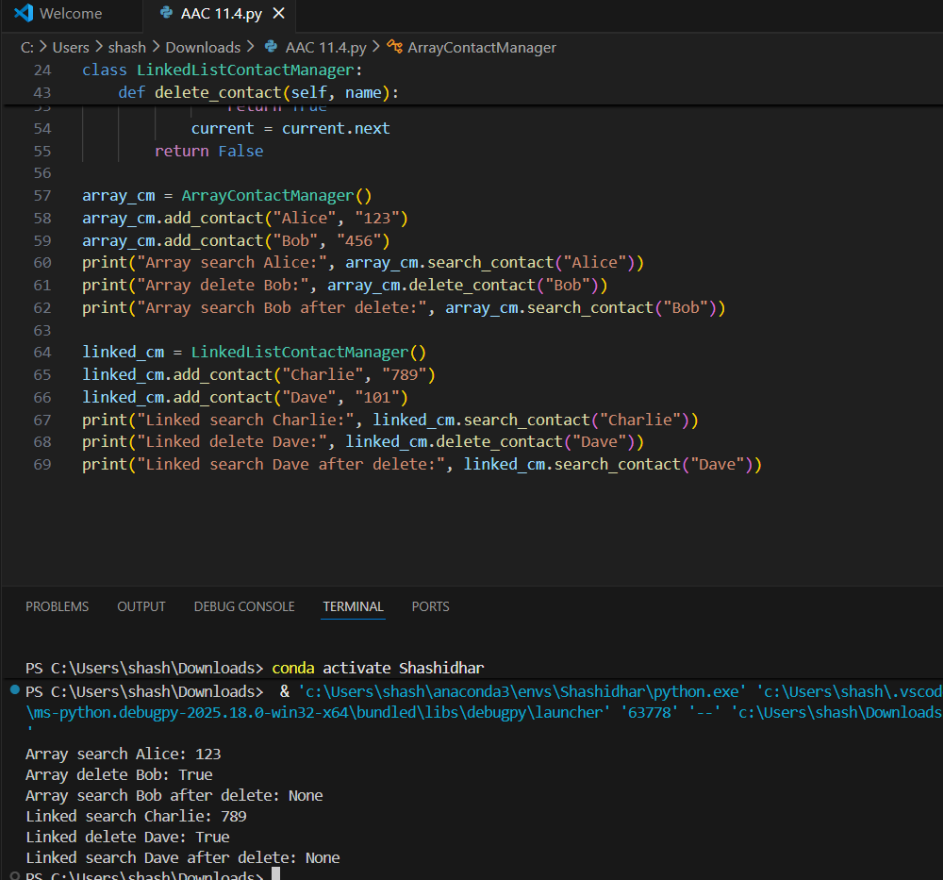
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

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PS C:\Users\shash\Downloads> conda activate Shashidhar
PS C:\Users\shash\Downloads> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63778' '--' 'c:\Users\shash\Downloads\AAC 11.4.py'
Array search Alice: 123
Array delete Bob: True
Array search Bob after delete: None
Linked search Charlie: 789
Linked delete Dave: True
Linked search Dave after delete: None
PS C:\Users\shash\Downloads>

```

	 <pre> C: > Users > shash > Downloads > AAC 11.4.py > ArrayContactManager 24 class LinkedListContactManager: 43 def delete_contact(self, name): 44 return True 54 current = current.next 55 return False 56 57 array_cm = ArrayContactManager() 58 array_cm.add_contact("Alice", "123") 59 array_cm.add_contact("Bob", "456") 60 print("Array search Alice:", array_cm.search_contact("Alice")) 61 print("Array delete Bob:", array_cm.delete_contact("Bob")) 62 print("Array search Bob after delete:", array_cm.search_contact("Bob")) 63 64 linked_cm = LinkedListContactManager() 65 linked_cm.add_contact("Charlie", "789") 66 linked_cm.add_contact("Dave", "101") 67 print("Linked search Charlie:", linked_cm.search_contact("Charlie")) 68 print("Linked delete Dave:", linked_cm.delete_contact("Dave")) 69 print("Linked search Dave after delete:", linked_cm.search_contact("Dave")) </pre> <pre> PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS PS C:\Users\shash\Downloads> conda activate Shashidhar PS C:\Users\shash\Downloads> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63778' '--' 'c:\Users\shash\Downloads\ Array search Alice: 123 Array delete Bob: True Array search Bob after delete: None Linked search Charlie: 789 Linked delete Dave: True Linked search Dave after delete: None PS C:\Users\shash\Downloads> </pre>	
	Task 2: Library Book Search System (Queues & Priority Queues) Scenario The SRU Library manages book borrow requests. Students and faculty submit requests, but faculty requests must be prioritized over student requests. Tasks 1. Implement a Queue (FIFO) to manage book requests. 2. Extend the system to a Priority Queue, prioritizing faculty requests. 3. Use GitHub Copilot to assist in generating: - o enqueue() method - o dequeue() method 4. Test the system with a mix of student and faculty requests. Expected Outcome • Working queue and priority queue implementations. • Correct prioritization of faculty requests.	

1from collections import deque
2import heapq
3class BookRequest:
4def __init__(self, requester_name, book_title, role):
5self.requester_name = requester_name
6self.book_title = book_title
7self.role = role
8def __repr__(self):
9return f"({self.role.upper()}) | ({self.requester_name}) -> '{self.book_title}'"
10class LibraryQueue:
11def __init__(self):
12self._queue = deque()
13def enqueue(self, request):
14self._queue.append(request)
15print(f"([QUEUE]) Added: {request}")
16def dequeue(self):
17if not self._queue:
18print("([QUEUE]) No requests in queue.")
19return None
20request = self._queue.popleft()
21print(f"([QUEUE]) Processing: {request}")
22return request
23def is_empty(self):
24return len(self._queue) == 0

PS C:\Users\shash\Downloads> cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda\envs\Shashidhar\python.exe' '
'c:\Users\shash\vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '59531' '-...' 'c:
:\Users\shash\Downloads\AAC_11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
PS C:\Users\shash\Downloads>

CHAT
LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION
Library Book Search System (Queues & Priority Queues)
Scenario
The SRU library manages book borrow requests. Students and faculty submit requests, but faculty requests must be prioritized over student requests.
Tasks
1. Implement a Queue (FIFO) to manage book requests.
2. Extend the system to a Priority Queue, prioritizing faculty requests.
3. Use GitHub Copilot to assist in generating:
o enqueue() method
o dequeue() method
4. Test the system with a mix of student and faculty requests.
Expected Outcome
• Working queue and priority queue implementations.
• Correct prioritization of faculty requests.
give code in python without comments just paste in the file

✓ Optimizing tool selection...
✓ Reviewed and created AAC_11.4.py
✓ Created + AAC_11.4.py
✓ Configured a Python environment
✓ Ran pylance-lsp-mcp-server - pylance mcp server (MCP Server)
✓ Reviewed code and provided a brief summary

+ AAC_11.4.py
Describe what to build next
Claude Sonnet 4.5

23def is_empty(self):
24return len(self._queue) == 0
25def display(self):
26if not self._queue:
27print("([QUEUE]) Queue is empty.")
28else:
29print("([QUEUE]) Current requests:")
30for req in self._queue:
31print(f" {req}")
32class PriorityLibraryQueue:
33PRIORITY_MAP = {"faculty": 0, "student": 1}
34
35def __init__(self):
36self._heap = []
37self._counter = 0
38def enqueue(self, request):
39priority = self.PRIORITY_MAP.get(request.role.lower(), 2)
40heapq.heappush(self._heap, (priority, self._counter, request))
41self._counter += 1
42print(f"([PRIORITY QUEUE]) Added: {request}")
43def dequeue(self):
44if not self._heap:

PS C:\Users\shash\Downloads> cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda\envs\Shashidhar\python.exe' '
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:\Users\shash\Downloads\AAC_11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
PS C:\Users\shash\Downloads>

CHAT
LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION
✓ Optimizing tool selection...
✓ Reviewed and created AAC_11.4.py
✓ Created + AAC_11.4.py
✓ Configured a Python environment
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✓ Reviewed code and provided a brief summary

Code written to AAC_11.4.py(c:\Users\shash\Downloads\AAC_11.4.py) and verified working. Here's what the output shows:

Regular Queue (FIFO): Processes in insertion order — Alice, Dr. Smith, Bob, Dr. Lee, Carol.

Priority Queue: Automatically reorders so faculty always come first — Dr. Smith, Dr. Lee (faculty), then Alice, Bob, Carol (students). Within the same role, arrival order (FIFO) is preserved.

Key classes:
• BookRequest — holds requester name, book title, and role
• LibraryQueue — FIFO using collections.deque with enqueue()/dequeue()
• PriorityLibraryQueue — min-heap via heapq assigning priority @ to faculty and @ to students

Claude Sonnet 4.5 - 1s

```

Welcome
AAC 11.4.py
C: > Users > shash > Downloads > AAC 11.4.py > ...
32 class PriorityLibraryQueue:
43     def dequeue(self):
45         print("[PRIORITY QUEUE] No requests in queue.")
46         return None
47         _, _ , request = heapq.heappop(self._heap)
48         print(f"[PRIORITY QUEUE] Processing: {request}")
49         return request
50     def is_empty(self):
51         return len(self._heap) == 0
52     def display(self):
53         if not self._heap:
54             print("[PRIORITY QUEUE] Queue is empty.")
55         else:
56             sorted_requests = sorted(self._heap, key=lambda x: (x[0], x[1]))
57             print("[PRIORITY QUEUE] Current requests (by priority):")
58             for _, _, req in sorted_requests:
59                 print(f"    {req}")
60 print("=" * 55)
61 print("          REGULAR QUEUE (FIFO) TEST")
62 print("=" * 55)
63 library_queue = LibraryQueue()
64 library_queue.enqueue(BookRequest("Alice", "Data Structures", "student"))
65 library_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\shash\Downloads> c::; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\S
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\laur
:\Users\shash\Downloads\AAC 11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
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[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
PS C:\Users\shash\Downloads>

```

```

C:\Users\shash\Downloads> AAC 11.4.py > ...
65 library_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))
66 library_queue.enqueue(BookRequest("Bob", "Operating Systems", "student"))
67 library_queue.enqueue(BookRequest("Dr. Lee", "Database Systems", "faculty"))
68 library_queue.enqueue(BookRequest("Carol", "Computer Networks", "student"))
69 print()
70 library_queue.display()
71 print()
72 print("--- Dequeuing all requests ---")
73 while not library_queue.is_empty():
74     library_queue.dequeue()
75 print()
76 print("=" * 55)
77 print("        PRIORITY QUEUE TEST (Faculty First)")
78 print("=" * 55)
79 priority_queue = PriorityLibraryQueue()
80 priority_queue.enqueue(BookRequest("Alice", "Data Structures", "student"))
81 priority_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))
82 priority_queue.enqueue(BookRequest("Bob", "Operating Systems", "student"))
83 priority_queue.enqueue(BookRequest("Dr. Lee", "Database Systems", "faculty"))
84 priority_queue.enqueue(BookRequest("Carol", "Computer Networks", "student"))
85 print()
86 priority_queue.display()
87 print()

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\shash\Downloads> c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy
:\Users\shash\Downloads\AAC 11.4.py'

```

```

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'

```

```
Welcome AAC 11.4.py
C: > Users > shash > Downloads > AAC 11.4.py > ...
76 print("=" * 55)
77 print("        PRIORITY QUEUE TEST (Faculty First)")
78 print("=" * 55)
79 priority_queue = PriorityLibraryQueue()
80 priority_queue.enqueue(BookRequest("Alice", "Data Structures", "student"))
81 priority_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))
82 priority_queue.enqueue(BookRequest("Bob", "Operating Systems", "student"))
83 priority_queue.enqueue(BookRequest("Dr. Lee", "Database Systems", "faculty"))
84 priority_queue.enqueue(BookRequest("Carol", "Computer Networks", "student"))
85 print()
86 priority_queue.display()
87 print()
88 print("--- Dequeuing all requests ---")
89 while not priority_queue.is_empty():
90     priority_queue.dequeue()
91

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\Downloads> c::; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anacond
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\deb
:\Users\shash\Downloads\AAC 11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
PS C:\Users\shash\Downloads> █
```

Task 3: Emergency Help Desk (Stack Implementation)

Scenario

SR University's IT Help Desk receives technical support tickets from students and staff. While tickets are received sequentially, issue escalation follows a **Last-In, First-Out (LIFO)** approach.

Tasks

1. Implement a **Stack** to manage support tickets.
2. Provide the following operations:
 - o push(ticket)
 - o pop()
 - o peek()
3. Simulate at least **five tickets** being raised and resolved.
4. Use **GitHub Copilot** to suggest additional stack operations such as:
 - o Checking whether the stack is empty
 - o Checking whether the stack is full (if applicable)

Expected Outcome

- Functional stack-based ticket management system.
- Clear demonstration of LIFO behavior.

WelcomeAAC 11.4py X

C:\Users> shash > Downloads > AAC 11.4py > ...
1 class SupportTicket:
2 def __init__(self, ticket_id, requester, issue, priority="normal"):
3 self.ticket_id = ticket_id
4 self.requester = requester
5 self.issue = issue
6 self.priority = priority
7
8 def __repr__(self):
9 return f"[#{self.ticket_id}] {self.requester} | {self.issue} ({self.priority.upper()})"
10
11
12 class HelpDeskStack:
13 MAX_SIZE = 10
14
15 def __init__(self):
16 self._stack = []
17
18 def push(self, ticket):
19 if self.is_full():
20 print(f"[STACK FULL] Cannot add: {ticket}")
21 return
22 self._stack.append(ticket)
23 print(f"[PUSHED] {ticket}")
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PS C:\Users\shash\Downloads> cd "c:\Users\shash\Downloads"; & "c:\Users\shash\anaconda\envs\shash\python.exe" "c:\Users\shash\vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher" "51957" "-..." "c:\Users\shash\Downloads\AAC 11.4.py"
SR UNIVERSITY IT HELP DESK - TICKET SYSTEM

--- Raising Support Tickets ---
[PUSHED] #101 Alice (Student) | Cannot access Wi-Fi (NORMAL)
[PUSHED] #102 Bob (Staff) | Projector not working in Lab3 (HIGH)
[PUSHED] #103 Carol (Student) | Email login failure (NORMAL)
[PUSHED] #104 Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python Debug Console

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION
o push(ticket)
o pop()
o peek()
3. Simulate at least five tickets being raised and resolved.
4. Use GitHub Copilot to suggest additional stack operations such as:
o Checking whether the stack is empty
o Checking whether the stack is full (if applicable)
Expected Outcome
• Functional stack-based ticket management system.
• Clear demonstration of LIFO behavior.
give code in python without comments

✓ Created AAC 11.5.py
✓ Created AAC 11.5.py
✓ Tried to run command but was blocked
✓ Ran pylance.runCodeInMcpServer - pylance mcp server (MCP Server)
✓ Summarized results for the user
Code saved to [AAC 11.5.py](C:\Users\shash\Downloads\AAC 11.5.py) and verified working. Here's what it demonstrates:
Classes:
• SupportTicket — holds ticket ID, requester, issue description, and priority
• HelpDeskStack — list-based stack with MAX_SIZE = 10
Operations Implemented:
Method Description
push(ticket) Adds ticket to top of stack
pop() Removes and resolves ticket from top (LIFO)
peek() Views top ticket without removing it
is_empty() Returns True if no tickets
is_full() Returns True if at MAX_SIZE
size() Returns current ticket count
display() Shows all tickets top-to-bottom

WelcomeAAC 11.4py X

C:\Users> shash > Downloads > AAC 11.4py > ...
12 class HelpDeskStack:
23 print(f"[PUSHED] {ticket}")
24
25 def pop(self):
26 if self.is_empty():
27 print(f"[STACK EMPTY] No tickets to resolve.")
28 return None
29 ticket = self._stack.pop()
30 print(f"[RESOLVED] {ticket}")
31 return ticket
32
33 def peek(self):
34 if self.is_empty():
35 print(f"[STACK EMPTY] No tickets in stack.")
36 return None
37 ticket = self._stack[-1]
38 print(f"[PEEK] Next to resolve: {ticket}")
39 return ticket
40
41 def is_empty(self):
42 return len(self._stack) == 0
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44 def is_full(self):
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PS C:\Users\shash\Downloads> cd "c:\Users\shash\Downloads"; & "c:\Users\shash\anaconda\envs\shash\python.exe" "c:\Users\shash\vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher" "51957" "-..." "c:\Users\shash\Downloads\AAC 11.4.py"
[PUSHED] #105 Eve (Student) | Printer offline in Library (NORMAL)
[STACK] Current tickets (top - most recent, total=5):
#105 Eve (Student) | Printer offline in Library (NORMAL)
#104 Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)
#103 Carol (Student) | Email login failure (NORMAL)
#102 Bob (Staff) | Projector not working in Lab3 (HIGH)
#101 Alice (Student) | Cannot access Wi-Fi (NORMAL)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python Debug Console

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION
✓ Created AAC 11.5.py
✓ Tried to run command but was blocked
✓ Ran pylance.runCodeInMcpServer - pylance mcp server (MCP Server)
✓ Summarized results for the user
Code saved to [AAC 11.5.py](C:\Users\shash\Downloads\AAC 11.5.py) and verified working. Here's what it demonstrates:
Classes:
• SupportTicket — holds ticket ID, requester, issue description, and priority
• HelpDeskStack — list-based stack with MAX_SIZE = 10
Operations Implemented:
Method Description
push(ticket) Adds ticket to top of stack
pop() Removes and resolves ticket from top (LIFO)
peek() Views top ticket without removing it
is_empty() Returns True if no tickets
is_full() Returns True if at MAX_SIZE
size() Returns current ticket count
display() Shows all tickets top-to-bottom

WelcomeAAC 11.4py X

C:\Users> shash > Downloads > AAC 11.4py > ...
12 class HelpDeskStack:
40
41 def is_empty(self):
42 return len(self._stack) == 0
43
44 def is_full(self):
45 return len(self._stack) >= self.MAX_SIZE
46
47 def size(self):
48 return len(self._stack)
49
50 def display(self):
51 if self.is_empty():
52 print(f"[STACK] Stack is empty.")
53 else:
54 print(f"[STACK] Current tickets (top - most recent, total={self.size}):")
55 for ticket in reversed(self._stack):
56 print(f" {ticket}")
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Welcome AAC 11.4.py X
C: > Users > shash > Downloads > AAC 11.4.py > ...
58
59 print("=" * 60)
60 print("      SR UNIVERSITY IT HELP DESK - TICKET SYSTEM")
61 print("=" * 60)
62
63 help_desk = HelpDeskStack()
64
65 print("\n--- Raising Support Tickets ---")
66 help_desk.push(SupportTicket(101, "Alice (Student)", "Cannot access Wi-Fi", "normal"))
67 help_desk.push(SupportTicket(102, "Bob (Staff)", "Projector not working in Lab3", "high"))
68 help_desk.push(SupportTicket(103, "Carol (Student)", "Email login failure", "normal"))
69 help_desk.push(SupportTicket(104, "Dr. Smith (Staff)", "VPN disconnecting repeatedly", "critical"))
70 help_desk.push(SupportTicket(105, "Eve (Student)", "Printer offline in Library", "normal"))
71
72 print()
73 help_desk.display()
74
75 print(f"\n[INFO] Stack empty? {help_desk.is_empty()}")
76 print(f"[INFO] Stack full? {help_desk.is_full()}")
77 print(f"[INFO] Total tickets in stack: {help_desk.size()}")
78
79 print("\n--- Peeking at top ticket ---")
80 help_desk.peek()
81
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\Downloads> c::; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '51957' '---'
:\Users\shash\Downloads\AAC 11.4.py'
--- Peeking at top ticket ---
[PEEK] Next to resolve: [#105] Eve (Student) | Printer offline in Library (NORMAL)

--- Resolving Tickets (LIFO Order) ---
[RESOLVED] [#105] Eve (Student) | Printer offline in Library (NORMAL)
[RESOLVED] [#104] Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)
[RESOLVED] [#103] Carol (Student) | Email login failure (NORMAL)
[RESOLVED] [#102] Bob (Staff) | Projector not working in Lab3 (HIGH)
```

```
C: > Users > shash > Downloads > AAC 11.4.py > ...
72     print()
73     help_desk.display()
74
75     print(f"\n[INFO] Stack empty? {help_desk.is_empty()}")
76     print(f"[INFO] Stack full? {help_desk.is_full()}")
77     print(f"[INFO] Total tickets in stack: {help_desk.size()}")
78
79     print("\n--- Peeking at top ticket ---")
80     help_desk.peak()
81
82     print("\n--- Resolving Tickets (LIFO Order) ---")
83     while not help_desk.is_empty():
84         help_desk.pop()
85
86     print()
87     print(f"[INFO] Stack empty after resolving all? {help_desk.is_empty()}")
88     help_desk.pop()
89
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads> c:: cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\an
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\lib
:\Users\shash\Downloads\AAC 11.4.py'
[RESOLVED] [#104] Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)
[RESOLVED] [#103] Carol (Student) | Email login failure (NORMAL)
[RESOLVED] [#102] Bob (Staff) | Projector not working in Lab3 (HIGH)
[RESOLVED] [#101] Alice (Student) | Cannot access Wi-Fi (NORMAL)

[INFO] Stack empty after resolving all? True
[STACK EMPTY] No tickets to resolve.
PS C:\Users\shash\Downloads>
```

Task 4: Hash Table

Objective

To implement a **Hash Table** and understand collision handling.

Task Description

Use AI to generate a hash table with:

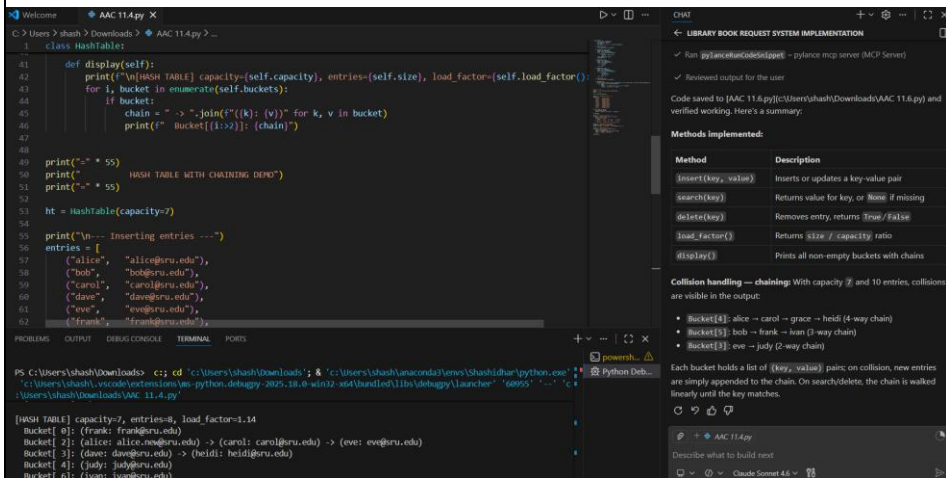
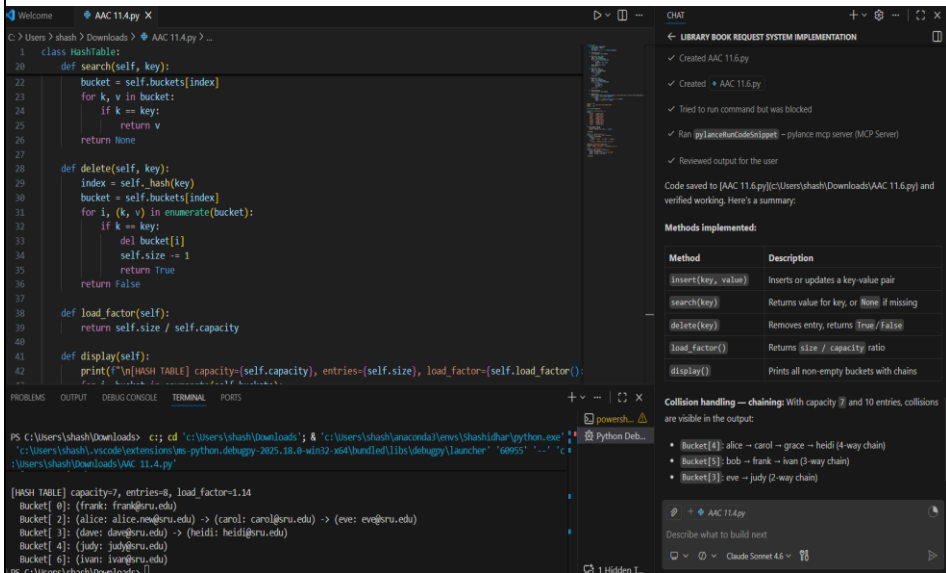
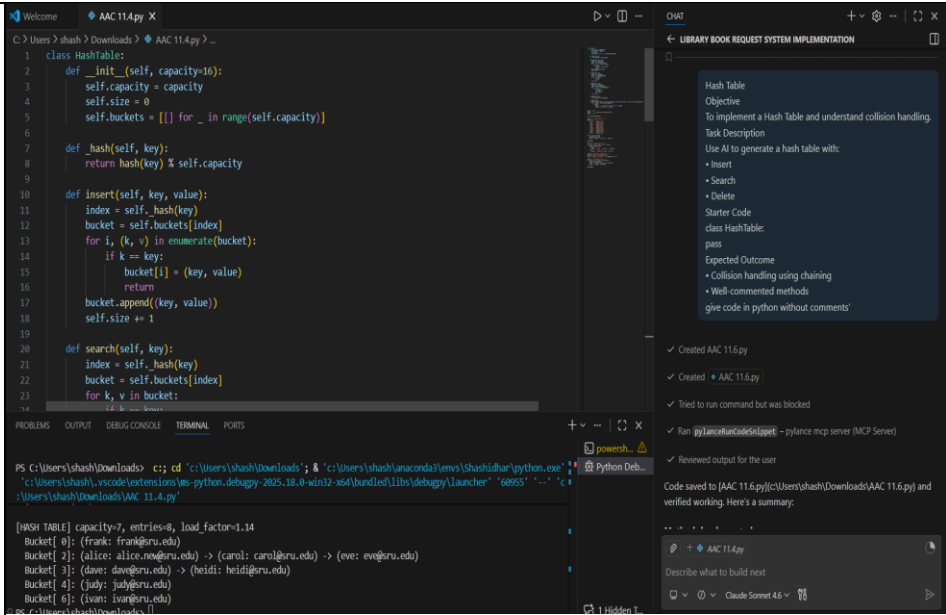
- Insert
- Search
- Delete

Starter Code

```
class HashTable:
    pass
```

Expected Outcome

- Collision handling using chaining
- Well-commented methods



C: > Users > shash > Downloads > AAC 11.4.py > ...

```
58     ("bob", "bob@sru.edu"),
59     ("carol", "carol@sru.edu"),
60     ("dave", "dave@sru.edu"),
61     ("eve", "eve@sru.edu"),
62     ("frank", "frank@sru.edu"),
63     ("grace", "grace@sru.edu"),
64     ("heidi", "heidi@sru.edu"),
65     ("ivan", "ivan@sru.edu"),
66     ("judy", "judy@sru.edu"),
67 ]
68 for key, value in entries:
69     ht.insert(key, value)
70     print(f" [INSERT] key='{key}' -> '{value}'")
71
72 ht.display()
73
74 print("\n--- Searching entries ---")
75 for key in ["alice", "eve", "judy", "unknown"]:
76     result = ht.search(key)
77     if result:
78         print(f" [FOUND] key='{key}' -> '{result}'")
79     else:
80         print(f" [MISSING] key='{key}' not found")
81
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\Downloads> c;; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\Scripts\python.exe' 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\debugpy_launcher.py' 'c:\Users\shash\Downloads\AAC 11.4.py'

```
[HASH TABLE] capacity=7, entries=8, load_factor=1.14
Bucket[ 0]: (frank: frank@sru.edu)
Bucket[ 2]: (alice: alice.new@sru.edu) -> (carol: carol@sru.edu) -> (eve: eve@sru.edu)
Bucket[ 3]: (dave: dave@sru.edu) -> (heidi: heidi@sru.edu)
Bucket[ 4]: (judy: judy@sru.edu)
Bucket[ 6]: (ivan: ivan@sru.edu)
```

```
C:\Users\shash> Downloads > AAC 11.4.py > ...
75 for key in ["alice", "eve", "judy", "unknown"]:
76     result = ht.search(key)
77     if result:
78         print(f" [FOUND] key='{key}' -> '{result}'")
79     else:
80         print(f" [MISSING] key='{key}' not found")
81
82 print("\n--- Updating an existing key ---")
83 ht.insert("alice", "alice.new@sru.edu")
84 print(f" [UPDATE] key='alice' -> '{ht.search('alice')}'")
85
86 print("\n--- Deleting entries ---")
87 for key in ["bob", "grace", "nonexistent"]:
88     success = ht.delete(key)
89     status = "DELETED" if success else "NOT FOUND"
90     print(f" [{status}] key='{key}'")
91
92 ht.display()
93
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads> c:: cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anacond
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\deb
:\Users\shash\Downloads\AAC 11.4.py'
```

```
[HASH TABLE] capacity=7, entries=8, load_factor=1.14
Bucket[ 0]: (frank: frank@sru.edu)
Bucket[ 2]: (alice: alice.new@sru.edu) -> (carol: carol@sru.edu) -> (eve: eve@sru.edu)
Bucket[ 3]: (dave: dave@sru.edu) -> (heidi: heidi@sru.edu)
Bucket[ 4]: (judy: judy@sru.edu)
Bucket[ 6]: (ivan: ivan@sru.edu)
PS C:\Users\shash\Downloads>
```

Task 5: Real-Time Application Challenge

Scenario

Design a **Campus Resource Management System** with the following features:

- Student Attendance Tracking
- Event Registration System
- Library Book Borrowing
- Bus Scheduling System
- Cafeteria Order Queue

Student Tasks

1. Choose the most appropriate data structure for each feature.
2. Justify your choice in **2–3 sentences**.
3. Implement **one selected feature** using AI-assisted code generation.

Expected Outcome

- Mapping table: Feature → Data Structure → Justification
- One fully working Python implementation

```
Welcome AAC 11.4.py AAC 11.7.py X
C:\Users\shash>Downloads> AAC 11.7.py > ...
1 from collections import deque
2 import heapq
3 from datetime import datetime
4
5 print("MAPPING: Feature -> Data Structure -> Justification")
6 print("-" * 60)
7 print("Attendance -> Dictionary -> O(1) Lookup by student ID")
8 print("Event Reg -> Set/Dictionary -> Deduplication + fast access")
9 print("Book Borrow -> Queue (FIFO) -> Serve requests in order")
10 print("Bus Schedule -> Priority Queue -> Earliest departure first")
11 print("Cafeteria -> Queue (FIFO) -> Orders prepared in sequence")
12
13 print("\n-- 1. Attendance (Dictionary) --")
14 attendance = {}
15 for sid, name, status in [("S001", "Alice", "present"), ("S002", "Bob", "absent"), ("S003", "Carol", "present"), ("S004", "Dave", "absent"), ("S005", "Eve", "present")]:
16     attendance.setdefault("2026-02-19", {})[sid] = (name, status)
17     print(f"({sid}) {name} -> {status}")
18 print("Report:", {sid: s for sid, (n, s) in attendance["2026-02-19"].items()})
19
20 print("\n-- 2. Event Registration (Set + Dict) --")
21 events = {"Tech Fest 2026": {"attendees": set(), "capacity": 5}}
22 for sid in ["S001", "S002", "S003", "S002", "S004", "S005", "S006"]:
23     e = events["Tech Fest 2026"]
24     if sid in e["attendees"] && len(e["attendees"]) >= e["capacity"]:
25         print(f" DUPLICATE: {sid}")
26     elif len(e["attendees"]) >= e["capacity"]:
27         print(f" FULL: {sid} rejected")
28     else:
29         e["attendees"].add(sid); print(f" Registered: {sid}")
30 print("Attendees:", sorted(events["Tech Fest 2026"]["attendees"]))
31
32 print("\n-- 3. Library Borrow (Queue) --")
33 bq = deque([("S001", "Data Structures"), ("S003", "Operating Systems"), ("S005", "Computer Networks"), ("S002", "Data Structures")])
34 while bq:
35     sid, book = bq.popleft()
36     print(f" Issued '{book}' to {sid}")
37
38 print("\n-- 4. Bus Scheduling (Priority Queue) --")
39 buses, ctr = [], 0
40 for route, t in [("City", "07:30"), ("Airport", "09:00"), ("Mall", "08:15"), ("Station", "07:00"), ("Hospital", "10:30")]:
41     heapq.heappush(buses, (datetime.strptime(t, "%H:%M"), ctr, route)); ctr += 1
42 while buses:
43     dt, c, route = heapq.heappop(buses)
44     print(f" Bus {c} at {dt} on route {route}")
45
46 print("\n-- 5. Cafeteria Orders (Queue) --")
47 orders = deque()
48 for i, order in enumerate(["Burger for Alice", "Pasta for Bob", "Sandwich for Carol", "Rice for Dave", "Noodles for Eve"]):
49     orders.append(i, order)
50 while orders:
51     i, order = orders.popleft()
52     print(f" Order #{i+1} ready: {order}")
```

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

Real-Time Application Challenge

Scenario

Design a Campus Resource Management System with the following features:

- Student Attendance Tracking
- Event Registration System
- Library Book Borrowing
- Bus Scheduling System
- Cafeteria Order Queue

Student Tasks

1. Choose the most appropriate data structure for each feature.
2. Justify your choice in 2-3 sentences.
3. Implement one selected feature using AI-assisted code generation.

Expected Outcome

- Mapping table: Feature -> Data Structure -> Justification
- One fully working Python implementation

give code in python without comments

Created AAC 11.7.py

Created AAC 11.7.py

Tried to run command to verify

shrink the code

Describe what to build next

Claude Sonnet 4.5

```
Welcome AAC 11.4.py AAC 11.7.py X
C:\Users\shash>Downloads> AAC 11.7.py > ...
17 print(f"({sid}) {name} -> {status}")
18 print("Report:", {sid: s for sid, (n, s) in attendance["2026-02-19"].items()})
19
20 print("\n-- 2. Event Registration (Set + Dict) --")
21 events = {"Tech Fest 2026": {"attendees": set(), "capacity": 5}}
22 for sid in ["S001", "S002", "S003", "S002", "S004", "S005", "S006"]:
23     e = events["Tech Fest 2026"]
24     if sid in e["attendees"] && len(e["attendees"]) >= e["capacity"]:
25         print(f" DUPLICATE: {sid}")
26     elif len(e["attendees"]) >= e["capacity"]:
27         print(f" FULL: {sid} rejected")
28     else:
29         e["attendees"].add(sid); print(f" Registered: {sid}")
30 print("Attendees:", sorted(events["Tech Fest 2026"]["attendees"]))
31
32 print("\n-- 3. Library Borrow (Queue) --")
33 bq = deque([("S001", "Data Structures"), ("S003", "Operating Systems"), ("S005", "Computer Networks"), ("S002", "Data Structures")])
34 while bq:
35     sid, book = bq.popleft()
36     print(f" Issued '{book}' to {sid}")
37
38 print("\n-- 4. Bus Scheduling (Priority Queue) --")
39 buses, ctr = [], 0
40 for route, t in [("City", "07:30"), ("Airport", "09:00"), ("Mall", "08:15"), ("Station", "07:00"), ("Hospital", "10:30")]:
41     heapq.heappush(buses, (datetime.strptime(t, "%H:%M"), ctr, route)); ctr += 1
42 while buses:
43     dt, c, route = heapq.heappop(buses)
44     print(f" Bus {c} at {dt} on route {route}")
45
46 print("\n-- 5. Cafeteria Orders (Queue) --")
47 orders = deque()
48 for i, order in enumerate(["Burger for Alice", "Pasta for Bob", "Sandwich for Carol", "Rice for Dave", "Noodles for Eve"]):
49     orders.append(i, order)
50 while orders:
51     i, order = orders.popleft()
52     print(f" Order #{i+1} ready: {order}")
```

```

Welcome  AAC 11.4.py  AAC 11.7.py X
C:\Users\shash> Downloads > AAC 11.7.py > ...
33 print(f" Issued '{book}' to {sid}")
34
35 print("\n-- 4. Bus Scheduling (Priority Queue) --")
36 buses, ctr = [], 0
37 for route, t in [("City", "07:30"), ("Airport", "09:00"), ("Mall", "08:15"), ("Station", "07:00"), ("Hospital", "10:30")
38 | heapq.heappush(buses, (datetime.strptime(t, "%H:%M"), ctr, route)); ctr += 1
39 while buses:
40     dt, _, r = heapq.heappop(buses)
41     print(f" Dispatch: {r} at {dt.strftime('%H:%M')}")
42
43 print("\n-- 5. Cafeteria Orders (Queue) --")
44 oq = deque(enumerate(["Alice:Burger", "Bob:Pasta", "Carol:Sandwich", "Dave:Rice", "Eve:Noodles"], 1))
45 while oq:
46     num, entry = oq.popleft()
47     name, item = entry.split(":")
48     print(f" Order #{num} ready: {item} for {name}")
49
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\shash\Downloads> c;; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50953' '--' 'c
:\Users\shash\Downloads\AAC 11.7.py'
MAPPING: Feature -> Data Structure -> Justification
-- 5. Cafeteria Orders (Queue) --
Order #1 ready: Burger for Alice
Order #2 ready: Pasta for Bob
Order #3 ready: Sandwich for Carol
Order #4 ready: Rice for Dave
Order #5 ready: Noodles for Eve
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.